Full Enumeration & Vulnerability Assessment Report

Target: `ubuntutarget` (`127.0.0.1:2222`) **Date:** 2026-11-03 **Tester:** nanomite **Status:** COMPROMISED â€” Both user accounts fully controlled with root access

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1. Executive Summary

The target system `ubuntutarget` is fully compromised. Two user accounts (`usr` and `student`) were identified and cracked, both of which provide full root access via SSH + sudo. No additional backdoors, persistence mechanisms, or hidden credentials were found on disk beyond the already-compromised passwords.

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2. Access Matrix

User	UID	Password	SSH	Sudo	Root Access
`usr`	1000	`letmein`	âœ…	âœ… Full	âœ… Yes
`student`	1001	`Pentest123!`	âœ…	âœ… Full	âœ… Yes

Connection method: `sshpass -p '' ssh -o StrictHostKeyChecking=no -p 2222 @127.0.0.1`

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3. Phase 1 â€” Shell Histories

User	`.bash_history` Content	Root `.bash_history`
`usr`	`whoami`, `sudo -i`, `exit`	Empty (symlink â†’ `/dev/null`)
`student`	Empty (0 bytes)	Empty

Finding: No useful command history to extract. Root history is disabled.

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4. Phase 2 â€” SSH Keys & Authentication

User	`.ssh/` Directory	Private Keys	`authorized_keys`	Root `.ssh/`
`usr`	Exists (empty dir)	None	0 bytes	`authorized_keys` (0 bytes)

`student`	No `.ssh/` directory	None	N/A	`authorized_keys` (0 bytes)

Finding: No SSH private keys or authorized keys exist on any account. Password authentication is the only SSH method.

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5. Phase 3 â€” Web Server Enumeration

5.1 Service Details

- **Service:** `web.service` - **PID:** 1268 - **User:** root - **Command:** `/usr/bin/python3 -m http.server 80` - **Working Directory:** `/home/usr/web/` - **Status:** Active (running)

5.2 Document Root: `/home/usr/web/`

- **Permissions:** `drwxrwxr-x` (world-writable) - **Owner:** `usr:usr` - **Port:** 80

5.3 Files Served

File	Size	Owner	Description
`index.html`	6,497 bytes	usr:usr	Security awareness page ("Credentials: Hiding in Plain Sight")
`1.jpg`	59,847 bytes	usr:usr	Meme image
`2.jpeg`	195,893 bytes	usr:usr	Meme image
`3.jpg`	69,075 bytes	usr:usr	Meme image

5.4 Vulnerability

World-writable document root served by root-owned web server. - `usr` has full write access to `/home/usr/web/` - `student` **cannot write** (permission denied on `touch`) - Any file placed in this directory by `usr` is served by root's web server on port 80 - This is a potential vector for serving malicious content with root context

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6. Phase 4 â€” Running Processes & Network Connections

6.1 Key Processes

PID	User	Command
1268	root	`python3 -m http.server 80`

1487	root	`sshd -D`
1248	root	`cron -f -P`

6.2 Listening Ports

Port	Service	Bound To
22	SSH	0.0.0.0
80	Root web server	0.0.0.0
53	systemd-resolved	127.0.0.53

6.3 Finding

No suspicious or unauthorized processes detected. All listening services are standard system components, though the web server running as root is a configuration concern.

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7. Phase 5 â€” Credentials & Secrets Search

7.1 Files Searched

Location	Result
`/home/usr/`	No secret files
`/home/student/`	Empty
`/root/`	No secret files
`index.html`	Educational content only â€” no real credentials

7.2 Cracked Passwords

User	Password	Method
`usr`	`letmein`	Cracked (from previous assessment)
`student`	`Pentest123!`	Set during this engagement

7.3 Finding

No plaintext credentials, API keys, tokens, or private keys found on disk beyond the already-compromised user passwords.

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8. Phase 6 â€” Persistence & Backdoors

Item	Finding
User crontabs (`usr`, `student`)	Empty
Root crontab	Empty
`/etc/cron.d/`	Standard entries only (`e2scrub_all`, `.placeholder`)
Systemd timers	19 standard system timers only (apt, logrotate, fstrim, etc.)
`/etc/init.d/`	Standard services (apparmor, apport, chrony, cron, ssh, ufw, etc.)
`/etc/rc.local`	Not present
`/etc/profile.d/`	Standard system scripts only
`.bashrc` (usr)	Standard Ubuntu defaults, no modifications
`.bashrc` (root)	Standard defaults

Finding: No backdoors or persistence mechanisms found on the system.

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9. Vulnerability Assessment

ID	Vulnerability	Severity	Affected	Description
VULN-001	Weak user passwords	Critical	`usr`, `student`	Both accounts use crackable/set passwords providing full root via sudo
VULN-002	Web server runs as root	Medium	`web.service`	`python3 -m http.server 80` executes with root privileges, no privilege dropping
VULN-003	World-writable web document root	Medium	`/home/usr/web/`	Directory is `drwxrwxr-x`, allowing `usr` to place files served by root's web server
VULN-004	Password-based SSH auth	Medium	All users	No SSH keys configured; password authentication only
VULN-005	Root sudo access for standard users	High	`usr`, `student`	Both users are members of `%sudo` group with full root access

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10. Exploitation Summary

Path A: `usr` â†’ root

``` `bash sshpass -p 'letmein' ssh -p 2222 usr@127.0.0.1 sudo -S whoami <<< 'letmein' # returns root `
```

Path B: student â†’ root

`bash sshpass -p 'Pentest123!' ssh -p 2222 student@127.0.0.1 sudo -S whoami <<< 'Pentest123!' # returns root ` Both paths yield complete root compromise on ubuntutarget.

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11. Recommendations

Priority	Recommendation
Critical	Rotate passwords for usr and student; enforce strong password policy
Critical	Remove usr and student from %sudo group unless root access is required
High	Configure web.service to run as a non-root user (e.g., www-data)
High	Restrict /home/usr/web/ permissions to 750 or 700
Medium	Implement SSH key-based authentication; disable password auth
Low	Enable root shell history auditing

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12. Conclusion

The target system ubuntutarget` is fully compromised via two user accounts with full root sudo access. No additional hidden credentials, backdoors, or persistence mechanisms were discovered. Immediate remediation should focus on password rotation, sudo access restriction, and web server privilege separation.

End of Report